PRACTICE QUESTION

1. Crisp Logic, Fuzzy Logic and First-Order Logic Inference

i. A smart-home system evaluates room temperature x ∈ [10, 40] °C. The system classifies temperatures as **Cold** or **Hot.** Demonstrate how these two categories can be represented using crisp and fuzzy logic.

ii. A restaurant evaluates food spiciness using a scale (x ∈ [0,10]).

The categories are:

- Mild
- Spicy

Demonstrate how these categories can be represented using crisp and fuzzy logic. Calculate the degree of membership for selected values of (x) and explain why fuzzy logic is more suitable when the boundary between mild and spicy is not clearly defined.

iii. Let two fuzzy sets be:

$A = \{0.2, 0.7, 0.8, 0.3\}, \quad B = \{0.7, 0.3, 0.5, 0.8\}$

Show the following fuzzy set operations
(i) A ∪ B (ii) A ∩ B (iii) A^c (iv) A-B (v) A is a subset of B

iv. Given:

1. Every programmer knows programming.
2. Every person who knows programming can solve programming problems.
3. Rahim is a programmer.

Represent the statements using FOL and prove that:

Rahim can solve programming problems.

Show the inference steps clearly.

v. Let fuzzy sets A and B over the same universe be represented by membership values:

Element	1	2	3	4
A	0.2	0.6	0.8	0.4
B	0.5	0.3	0.7	0.9

Using standard fuzzy operations, calculate A ∪ B, A ∩ B and A^c. Then determine whether B is a fuzzy subset of A.

2. K-means Classifier, Linear Regression and Logistic Regression

i. Six customers are represented by two features (Age, Spending Score): P1=(1,1), P2=(2,2), P3=(3,1), P4=(8,8), P5=(9,7), P6=(8,9). For K=2, the initial centroids are C1=(2,1) and C2=(8,8).

a. Calculate the Euclidean distance of each point from both centroids. b. Assign each point to the nearest cluster. c. Calculate the updated centroid of each cluster after one iteration.

ii. The following data describe study hours x and exam score y:

Study hours x	**1**	**2**	**3**	**4**
Exam score y	2	3	5	4

Using the linear regression model $\hat{y} = b_0 + b_1 x$, calculate b_0 and b_1, then predict the score for x=5 hours.

iii. A logistic regression model predicts whether a machine will fail. The linear predictor is $v = 0.54T - 12$, where T is temperature in °C, and the activation function is sigmoid.

Calculate P(Fail=1) and P(Fail=0) at T=60°C. ii. Using a 0.5 threshold, give the classification. iii. Find the temperature at which P(Fail=1)=0.73.

3. Perceptron Algorithm and Artificial Neural Networks

i. A perceptron has inputs $x_1=1$ and $x_2=0$, weights $w_1=0.2$ and $w_2=-0.1$, bias $b=-0.1$, learning rate $\eta=0.2$, and a it uses sigmoid activation. The target output is 0.

Calculate the output y after one forward propagation. If the prediction is incorrect, update w_1, w_2 and b using $w_i(new)=w_i+\eta\delta x_i$ and $b(new)=b+\eta\delta$.

Here, δ= y * (1 - y) * (Target_Value - y)

ii. A perceptron has two layers. For first layer the inputs are $x_1=2$ and $x_2=4$, weights $w_1=0.1$ and $w_2=0.3$, bias b=.1, and then this output goes to the second layer as input with weight $w_3=0.1$ and bias b=0. Both the layers use sigmoid activation function on the weighted sum of the inputs.

a. Perform one forward propagation: calculate the hidden neuron output and then the final output. Give the final answer to three decimal places.

b. If the target is 5, state whether an update is required. If an update is required, explain why.

4. Monte Carlo Simulation and Random Search

i. A delivery company records the number of packages delivered per day and their frequencies:

Packages/day	**20**	**30**	**40**	**50**
Frequency	10	20	40	30

The following random numbers are generated: 07, 34, 58, 91, 42, 16, 27. Use two-digit random-number intervals based on the frequency distribution to simulate the seven days. Calculate the estimated average number of packages delivered per day.

ii. A Monte Carlo simulation estimates the average waiting time of customers. With 20 random observations, the estimated average is 8.4 minutes. With 2,000 random observations, the estimated average is 8.1 minutes.

Which estimate would generally be considered more reliable? ii. Briefly explain why increasing the number of random observations usually improves the simulation estimate.

5. Markov Model

A music streaming service tracks whether a listener's **daily mood** is **Happy (H)** or **Calm (C)**. Based on historical data the transition matrix is:

$$P = \begin{bmatrix} 0.75 & 0.25 \\ 0.40 & 0.60 \end{bmatrix}$$

If the listener continues following this pattern for a long period of time, what proportion of days will the listener be **Happy** and what proportion will be **Calm**? Calculate the **long-run (stationary) distribution** from the given transition matrix.